

# Graph Class - 2

Special class

adj Matrix

└ T.C  $\rightarrow O(V^2)$   
└ S.C  $\rightarrow O(V^2)$

adj List

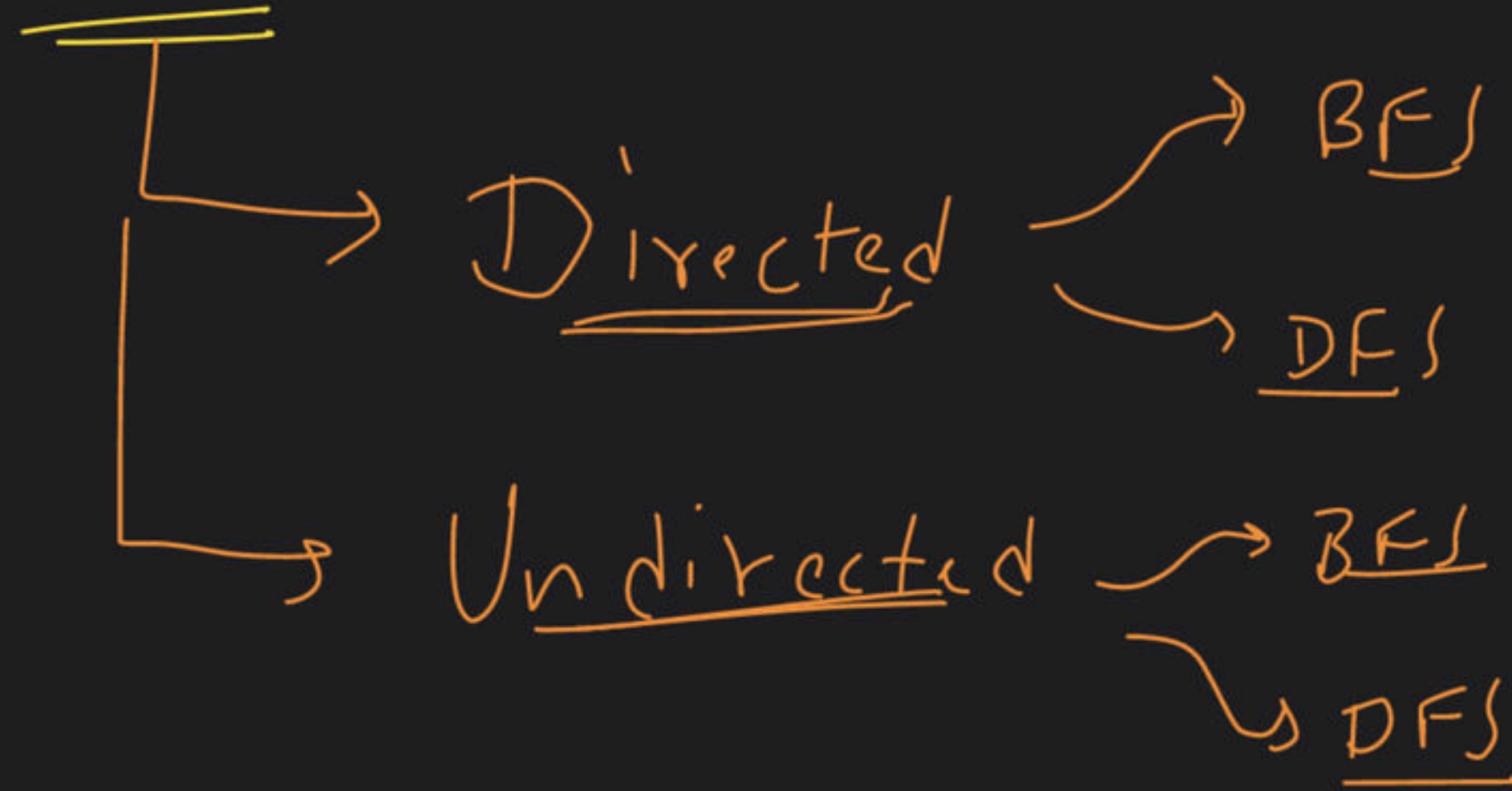
└ T.C  $\rightarrow O(V + E)$   
└ S.C  $\rightarrow O(V + E)$

no. of nodes  
└  $V$

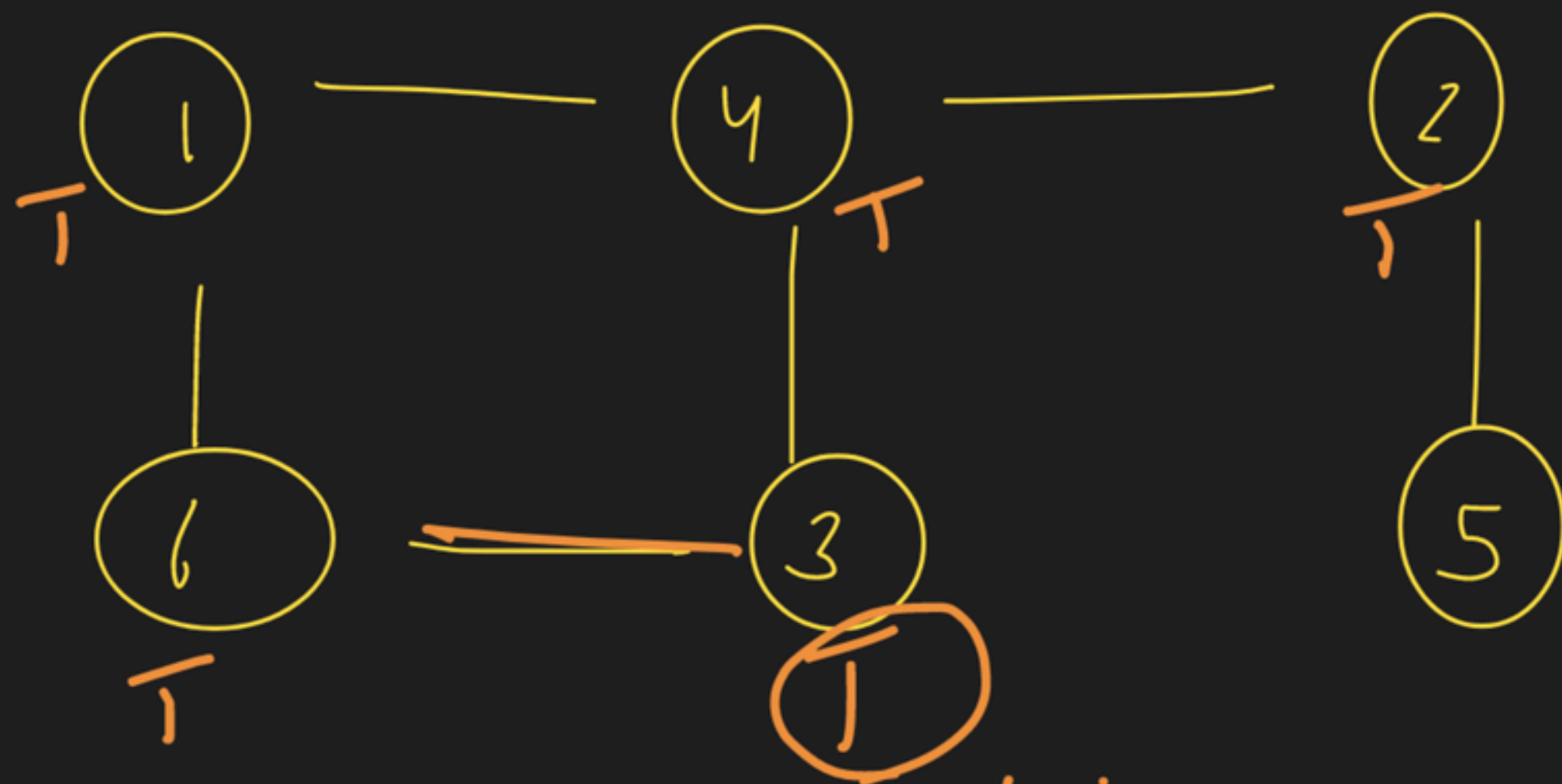
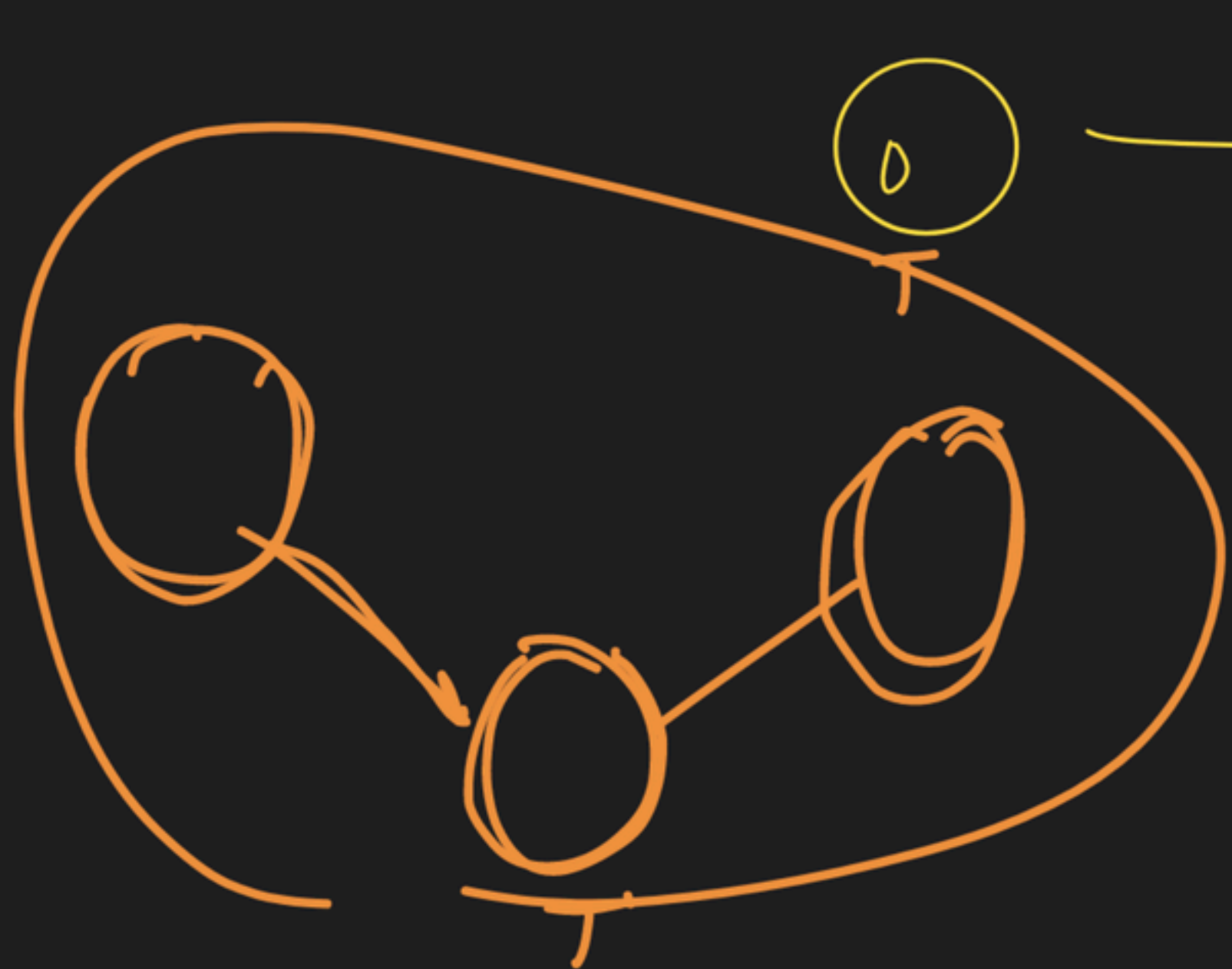
BFS / DFS

└ T.C  $\rightarrow O(V + E)$   
S.C  $\rightarrow O(V)$

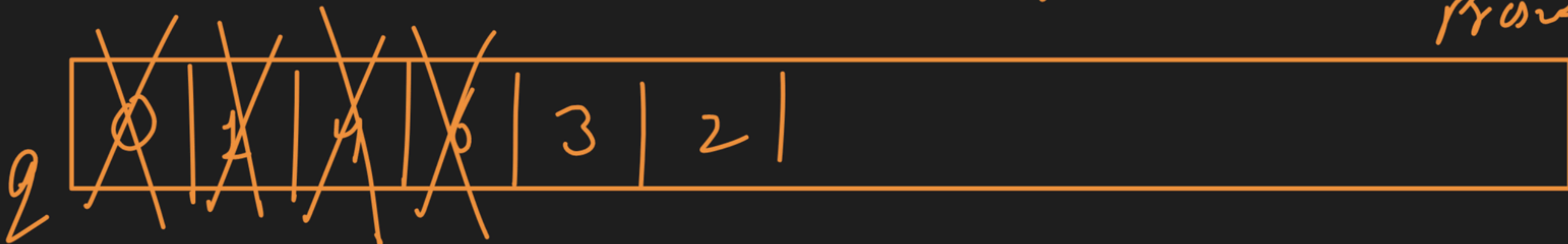
# → Cycle Detection

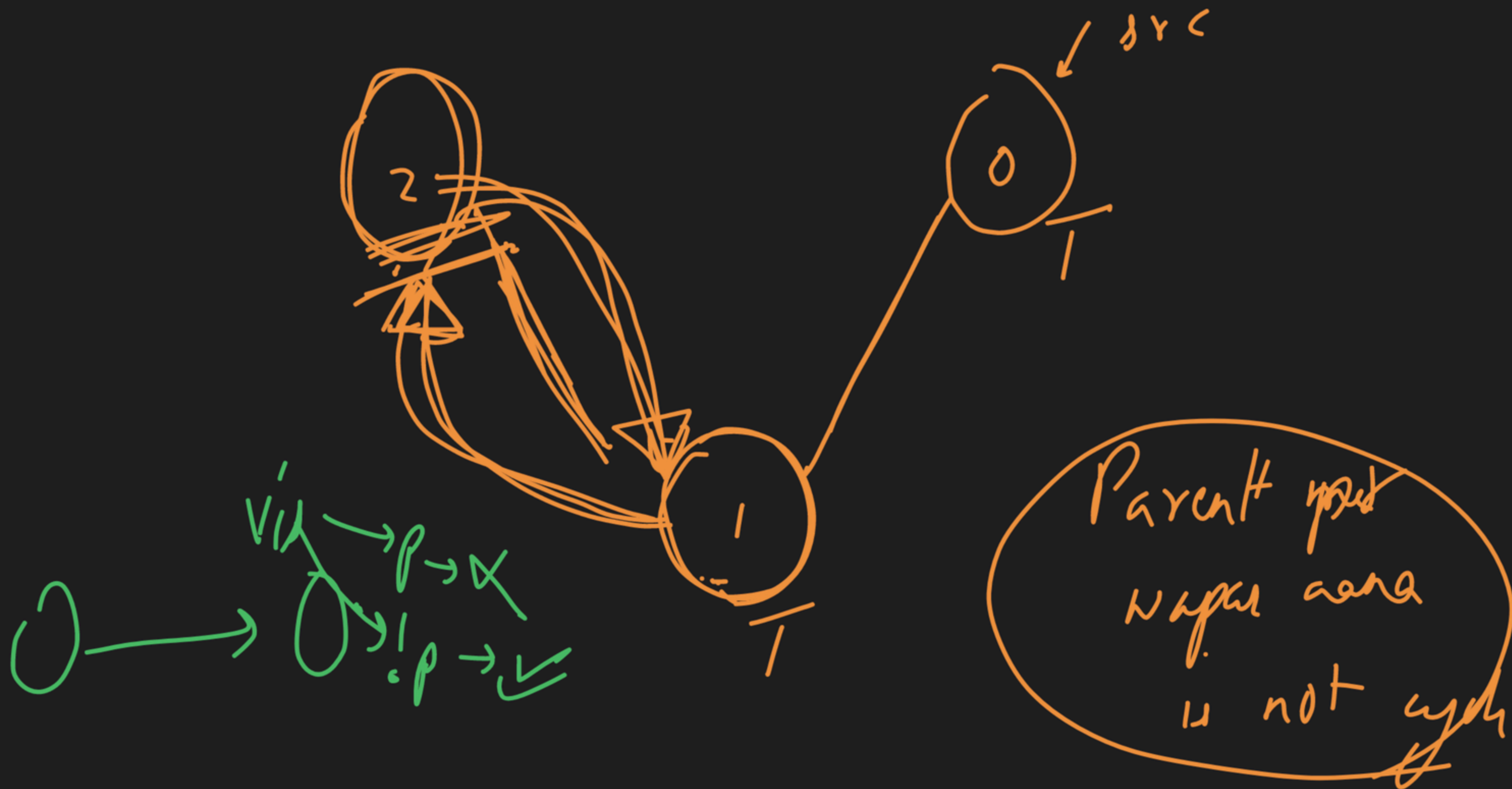


① Undirected → BFS

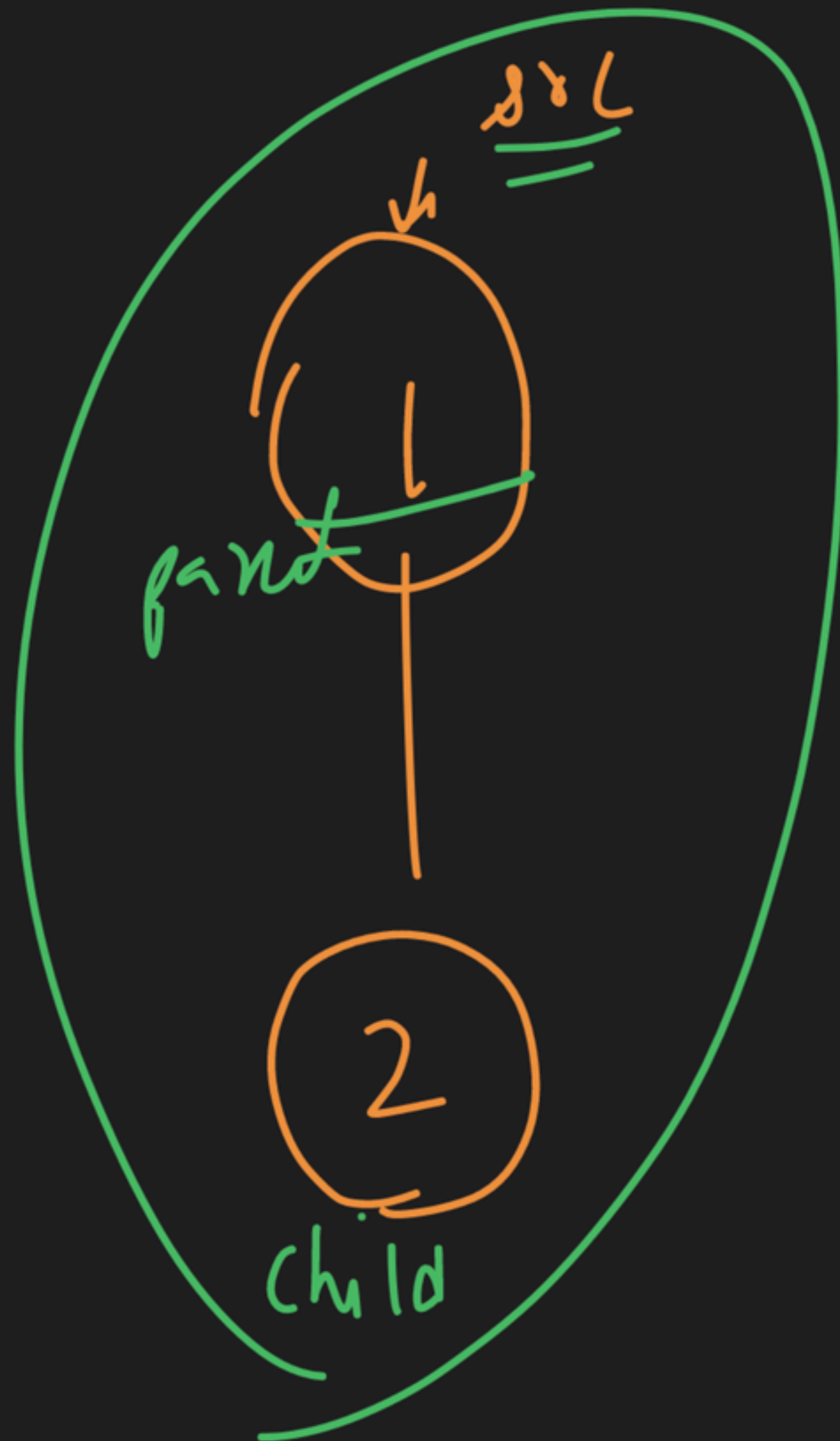


nbr → already visited → cycle found

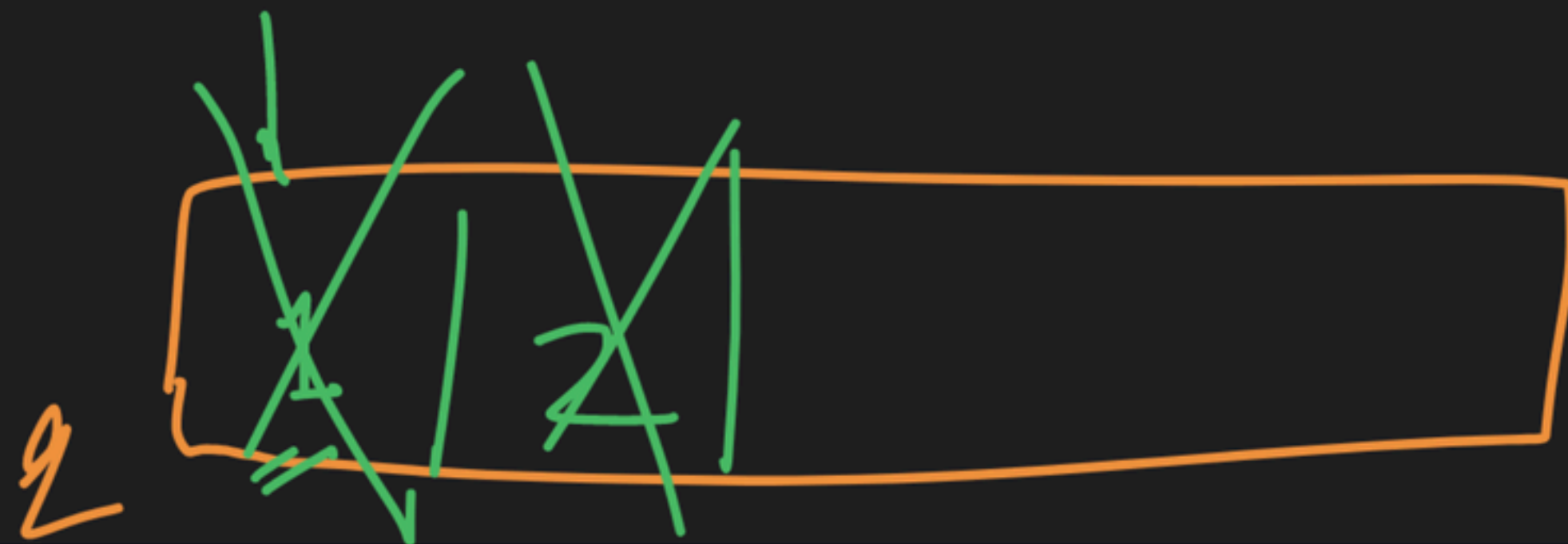


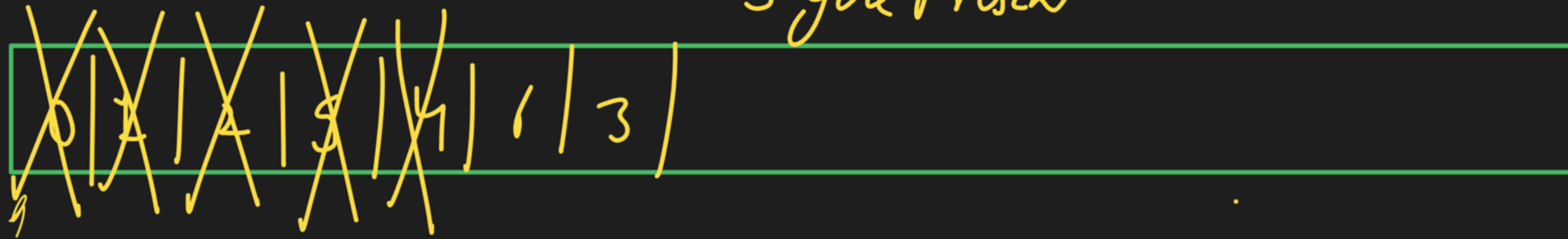
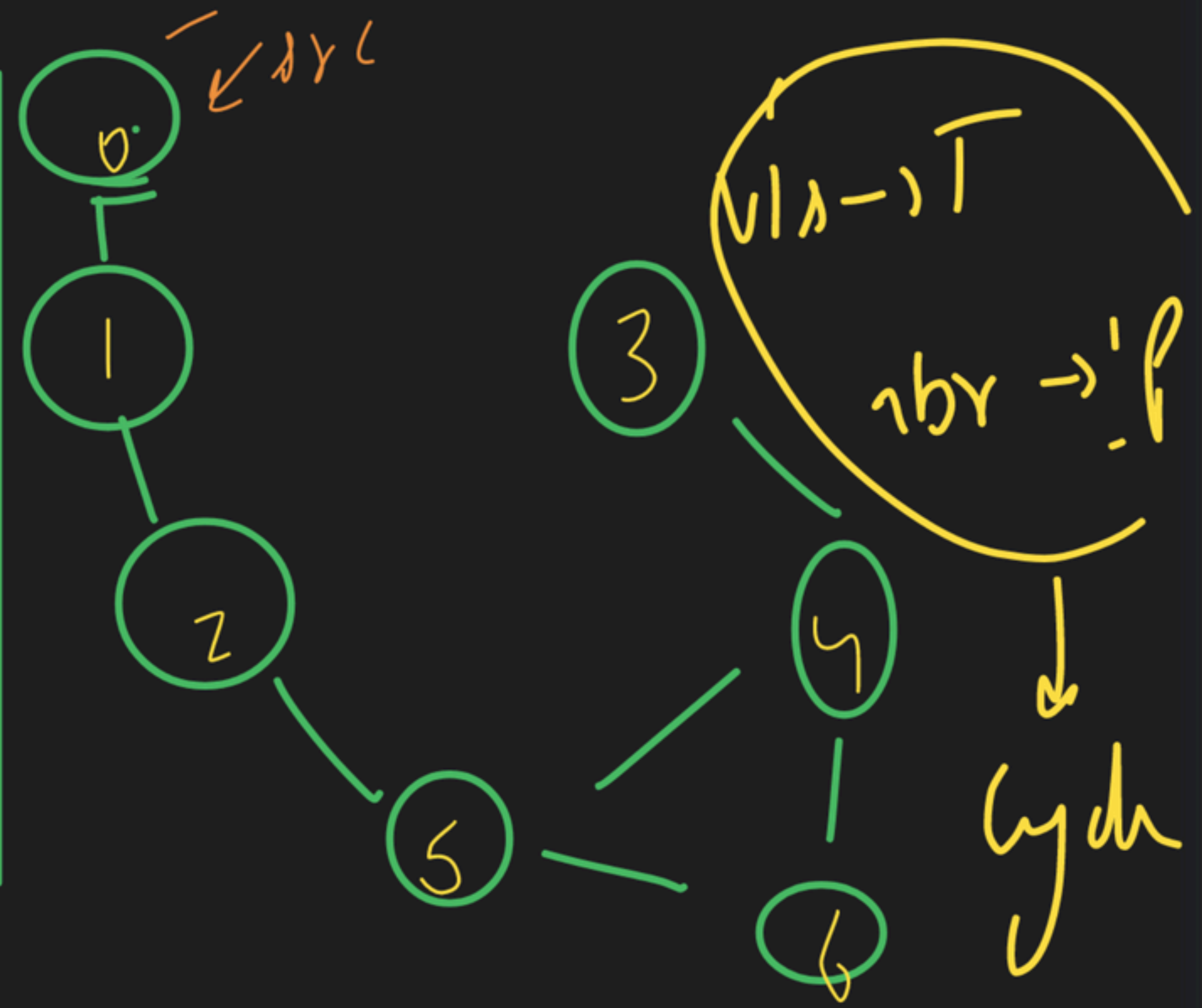
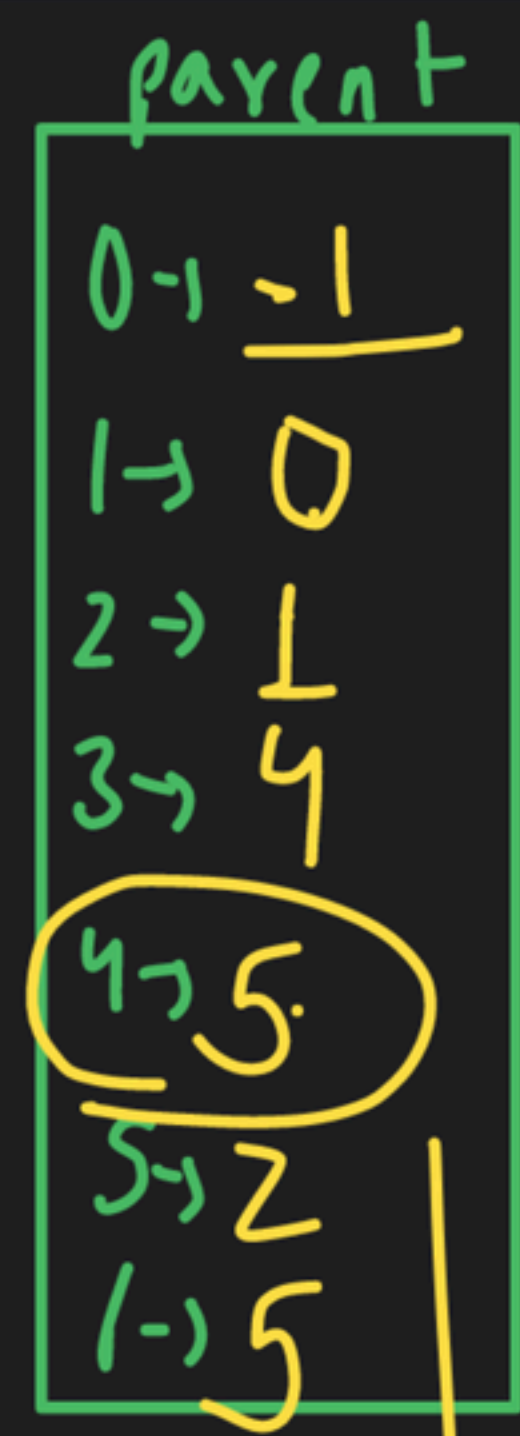
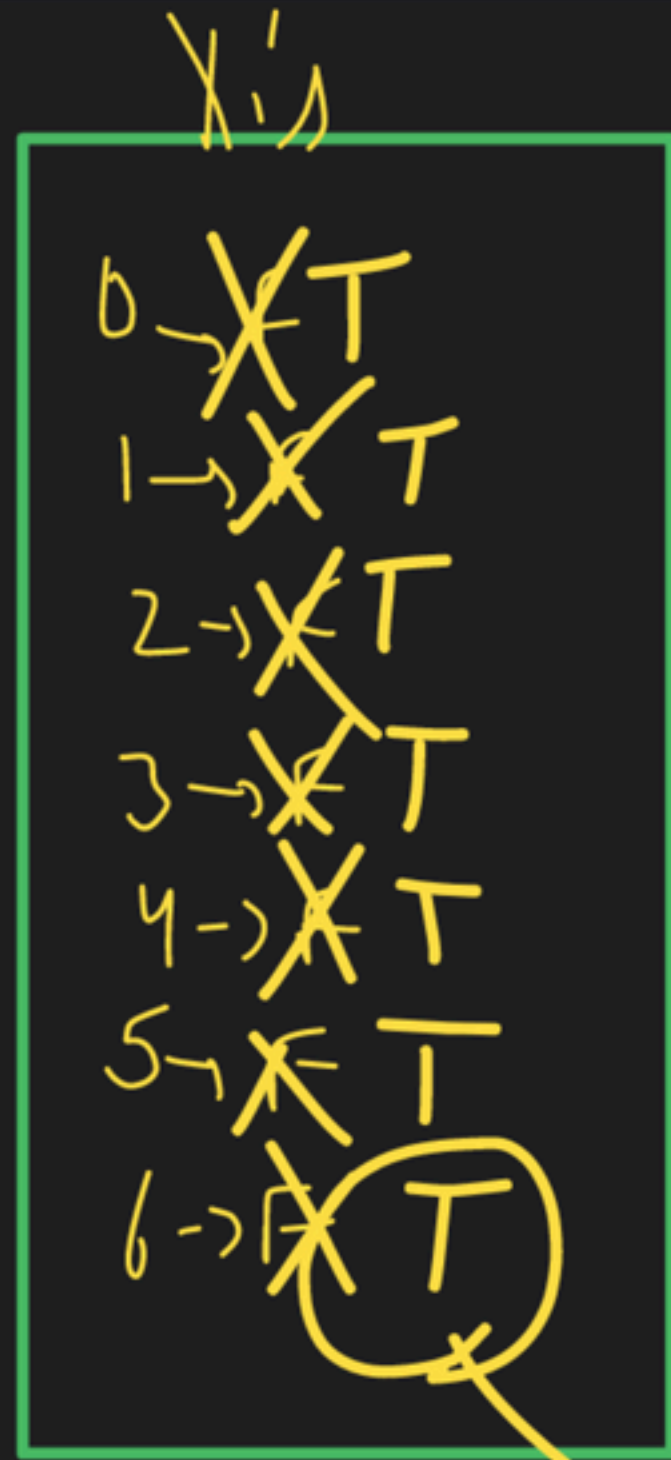
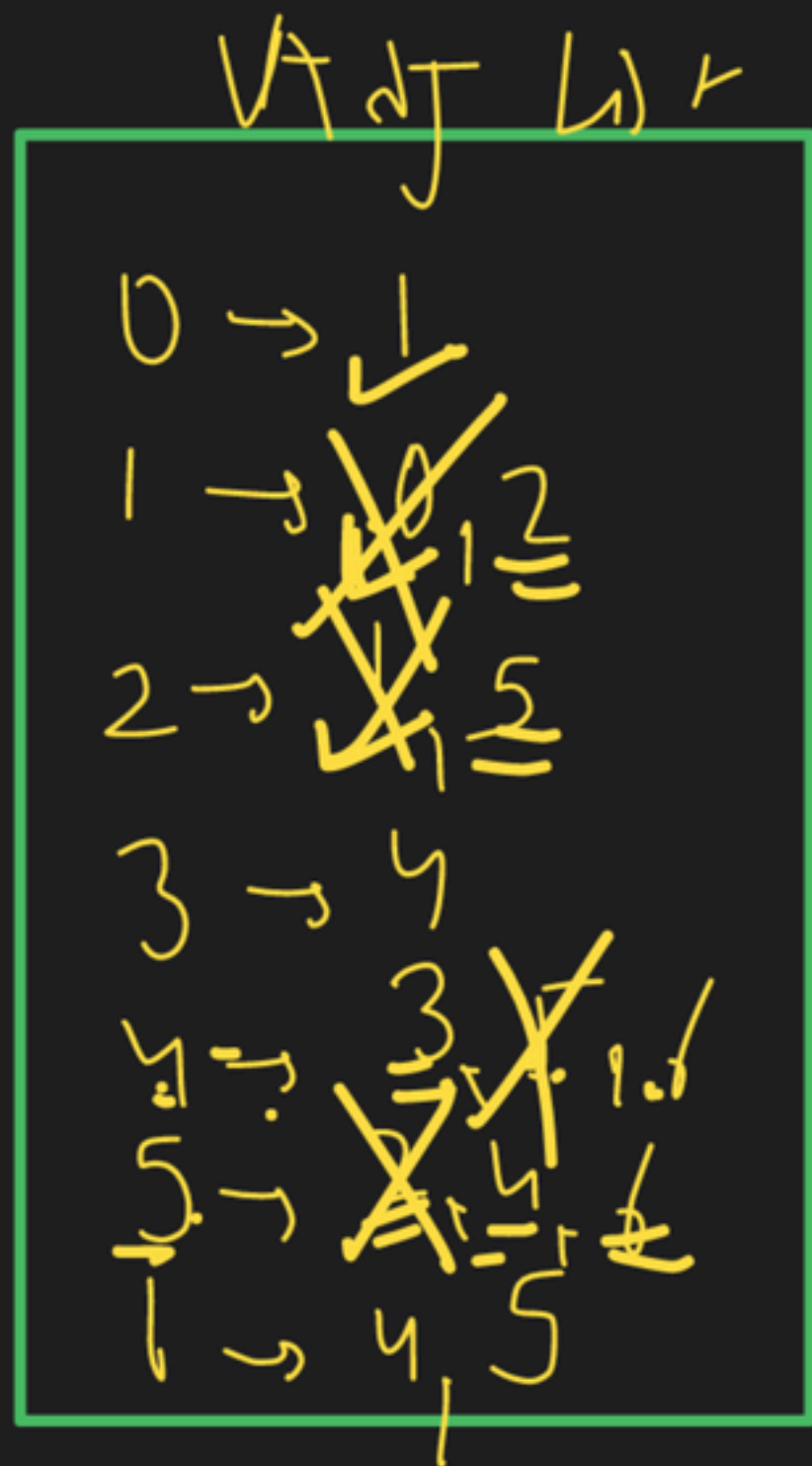


1  $\rightarrow$  ~~F~~ T  
2  $\rightarrow$  ~~F~~ T



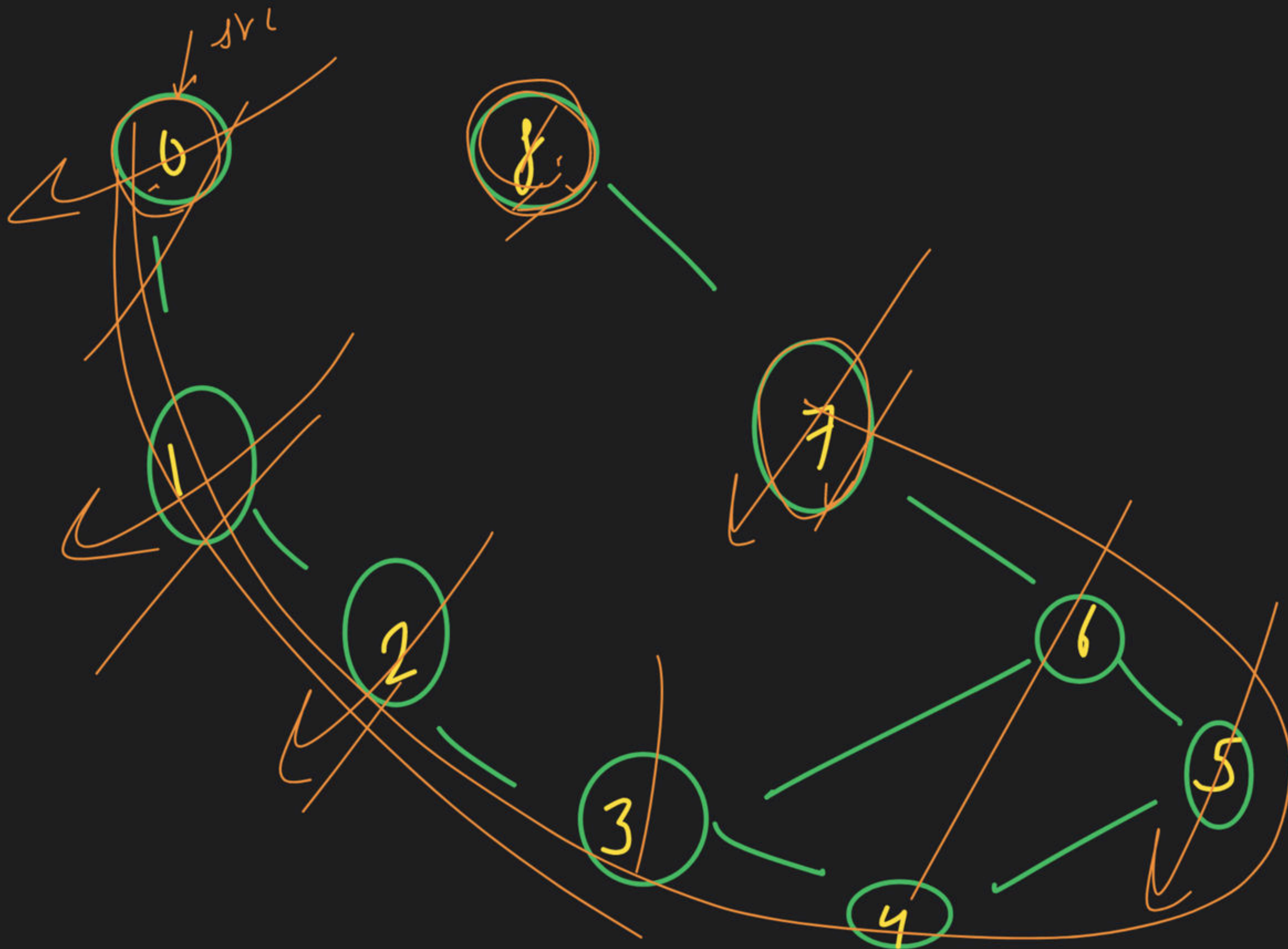
1  $\rightarrow$  {2}  
2  $\rightarrow$  {1}

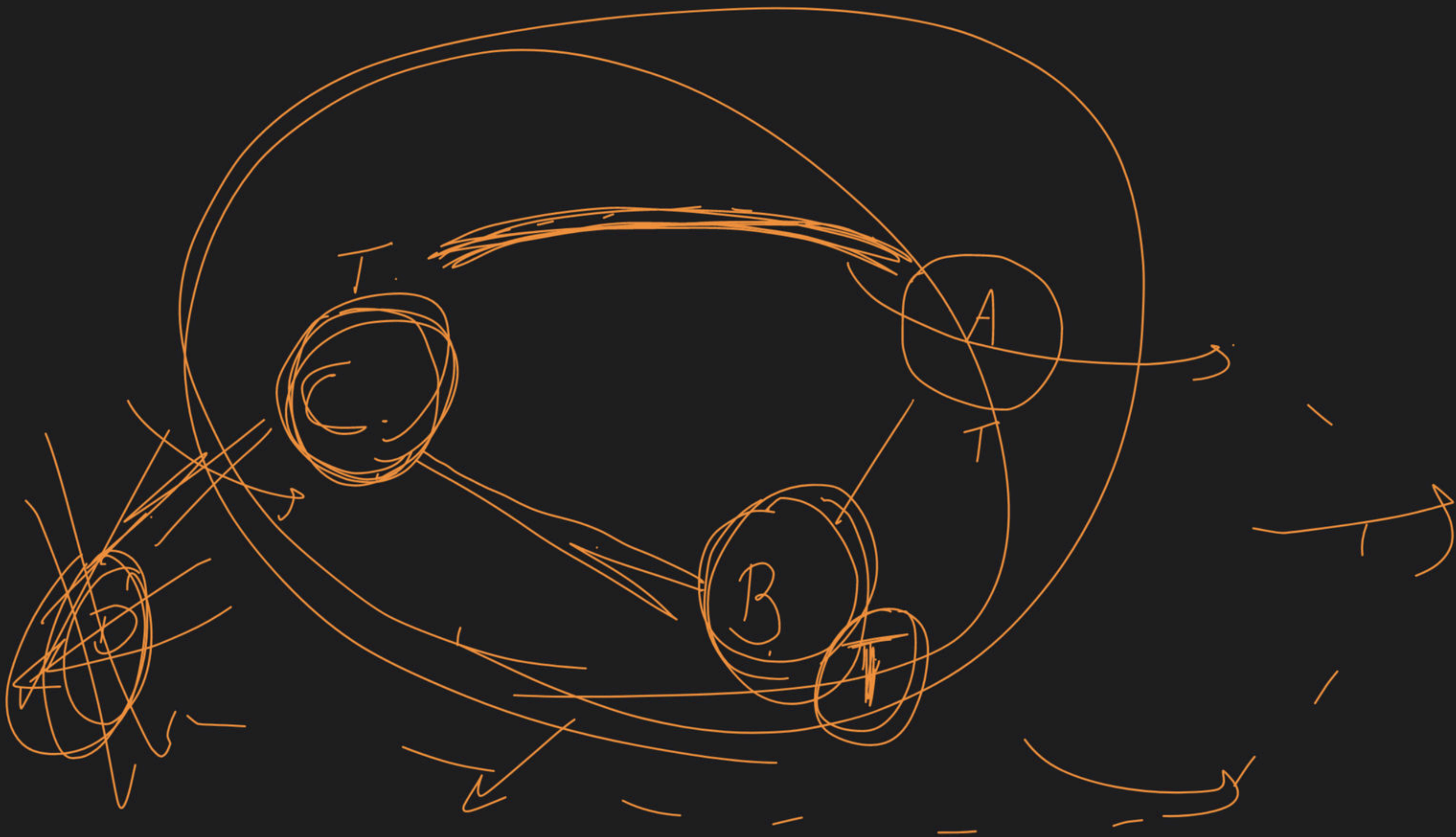


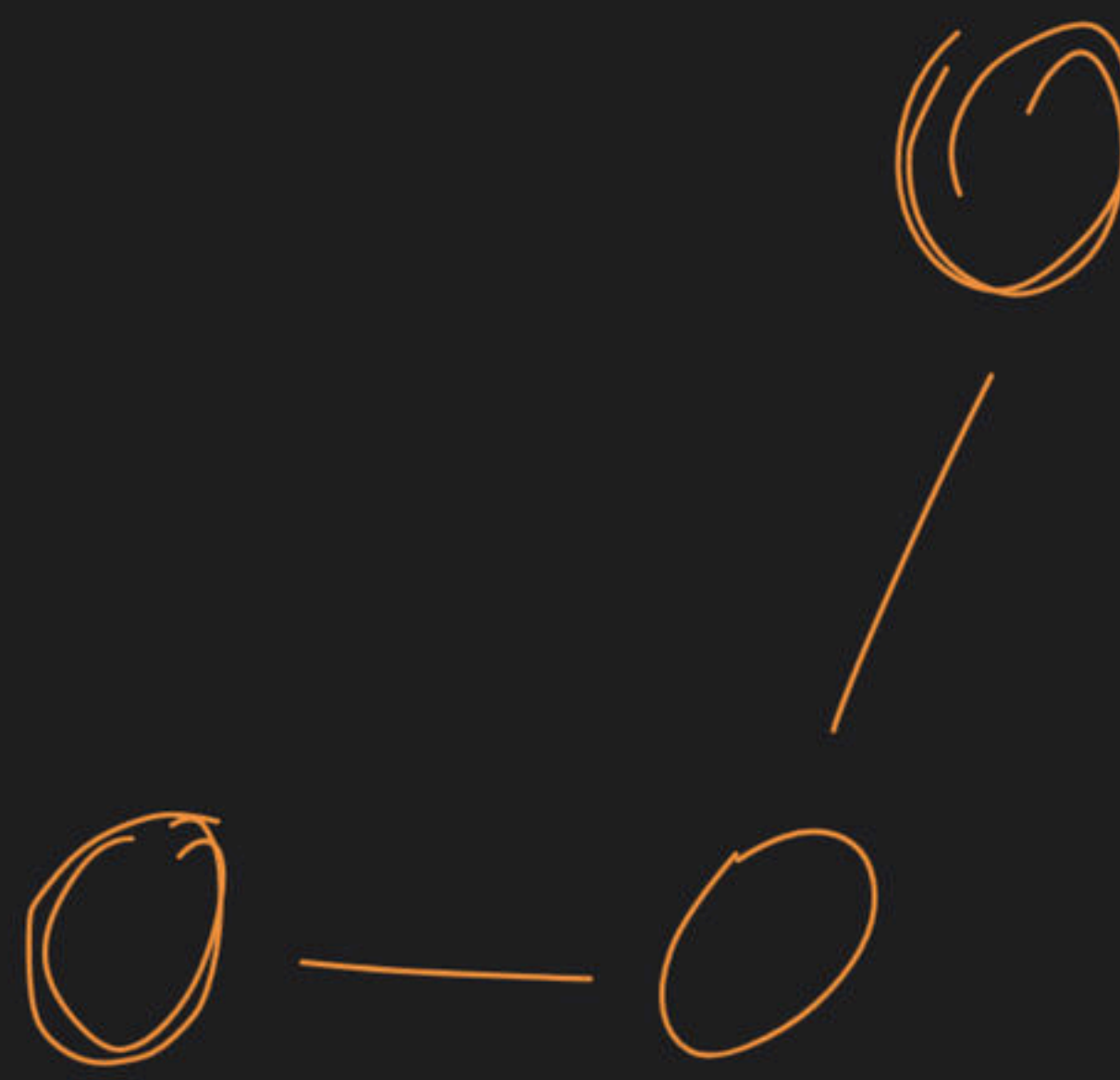
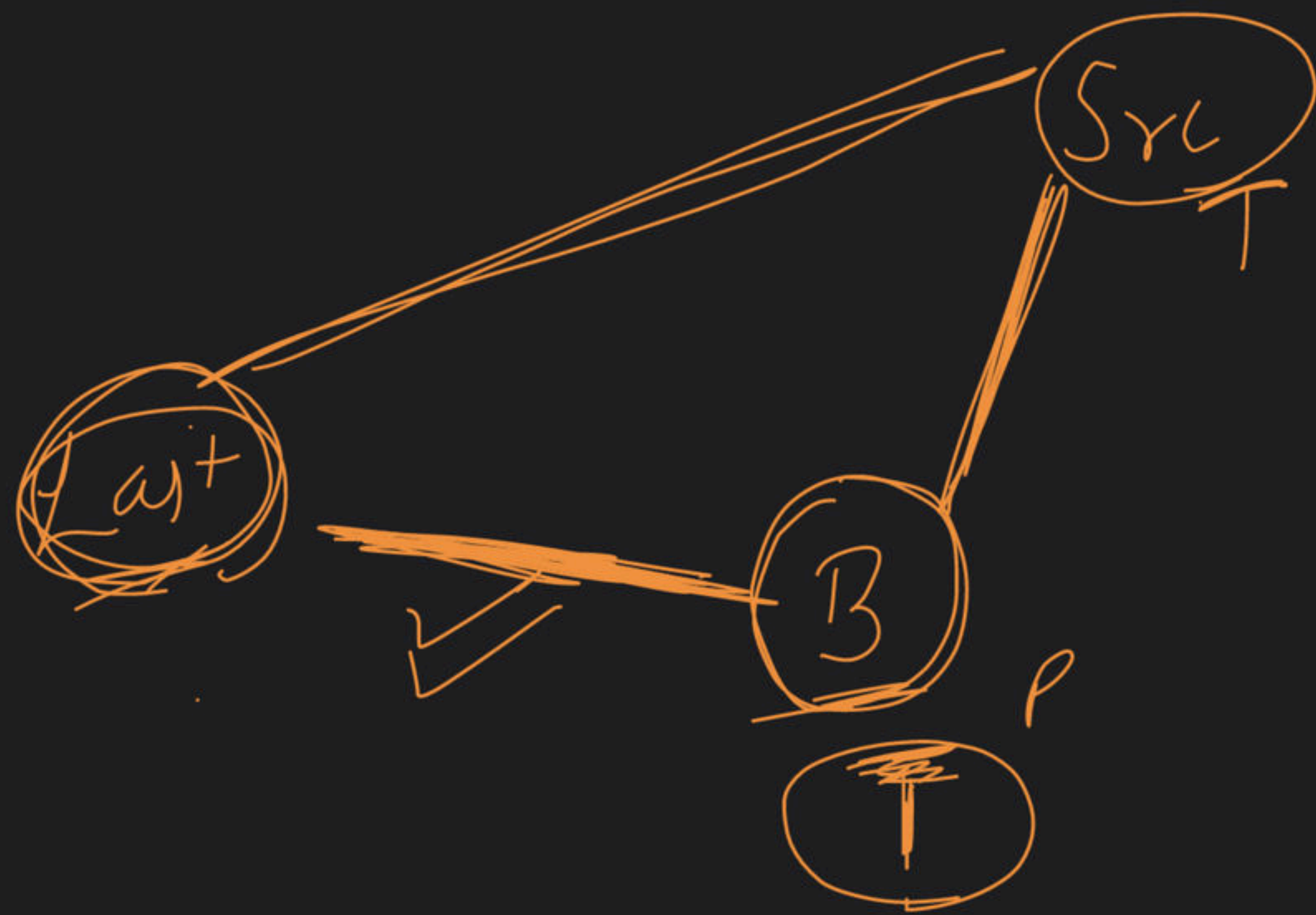


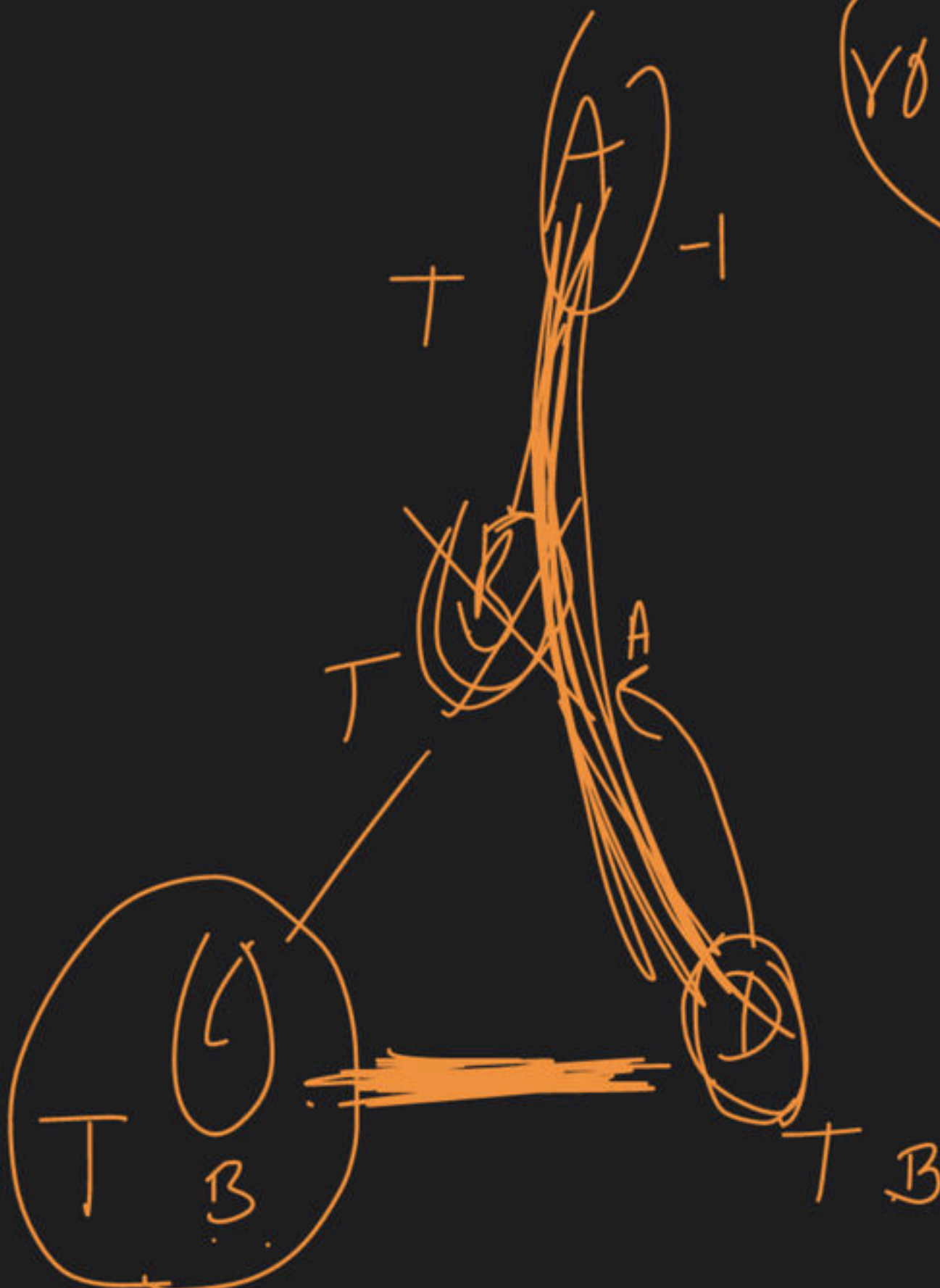
cycle present

Bf

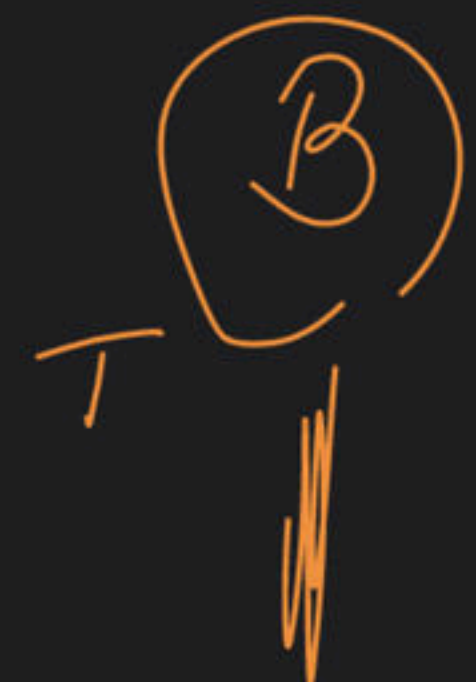








rotten  
tomatoes



for (

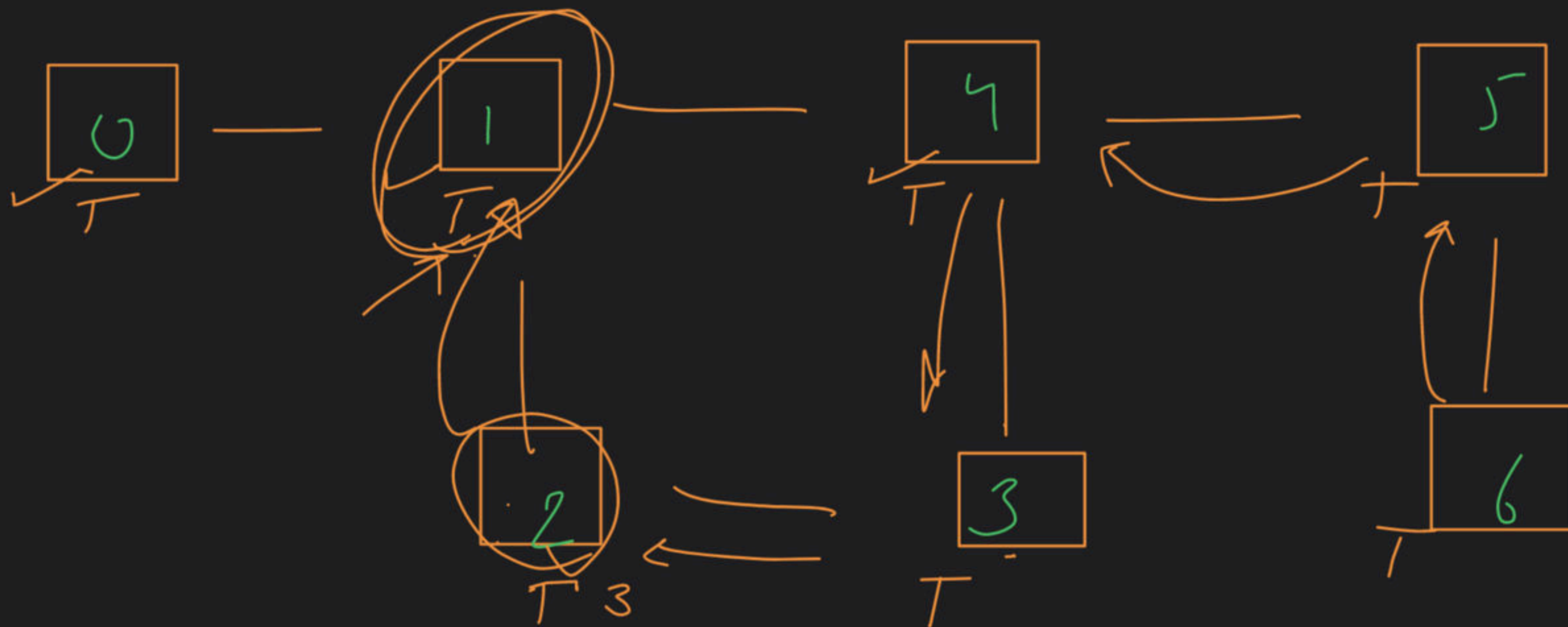
2 Hw

find the number  
of islands

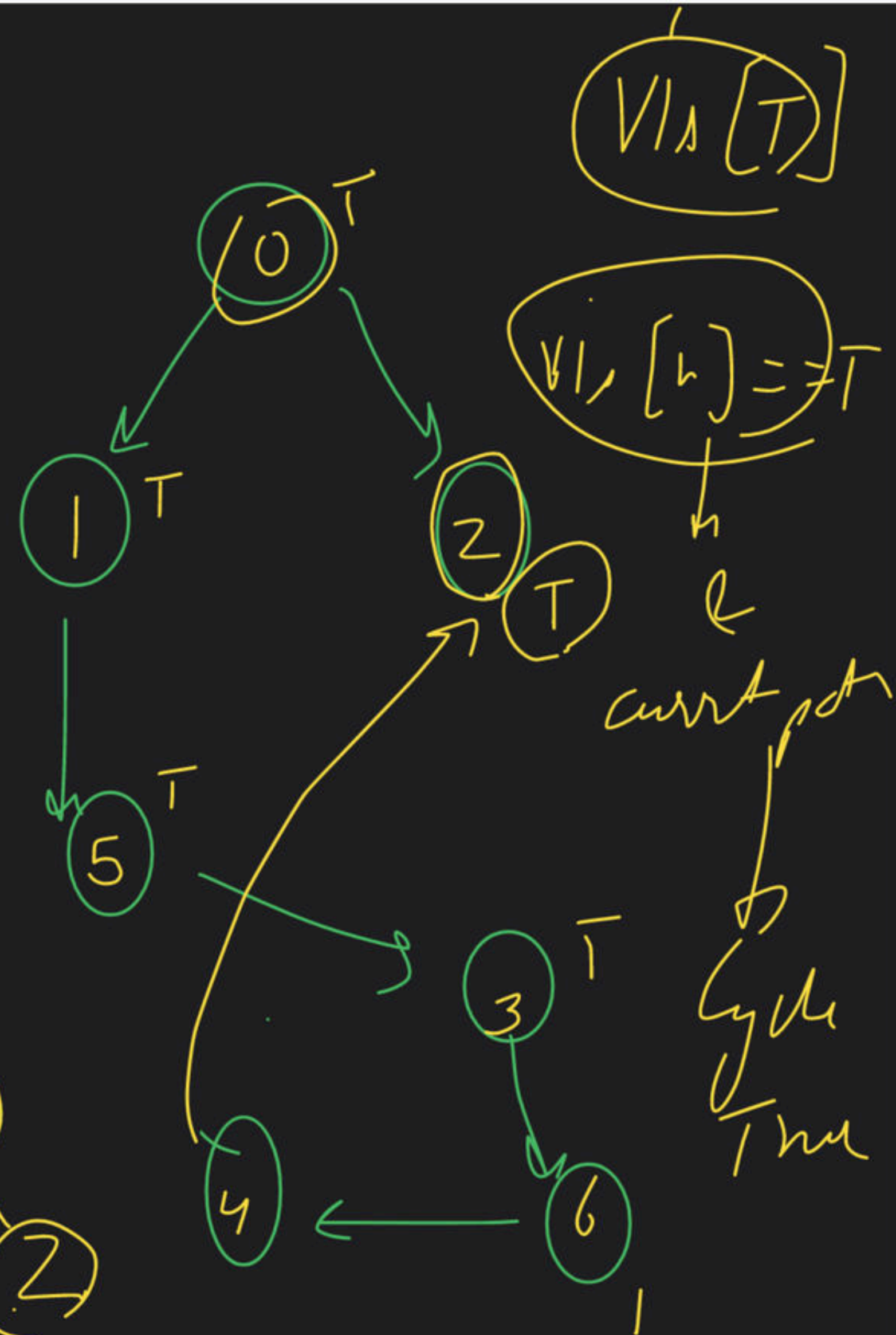
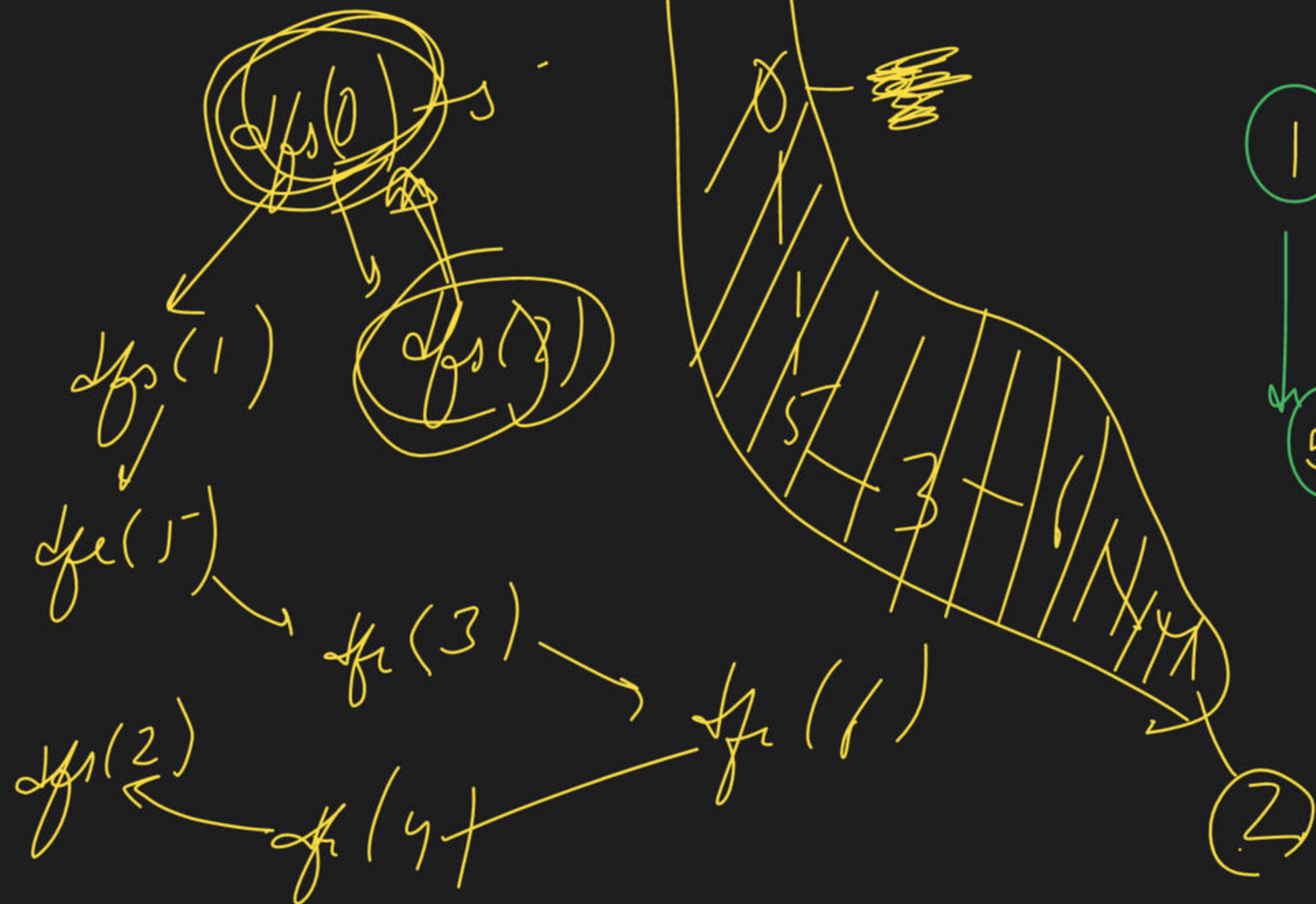
```

for (
{
  if (!vis)
  {
    → recu
    count++
  }
}
  
```

Vrdivected graph  $\rightarrow$  DFS



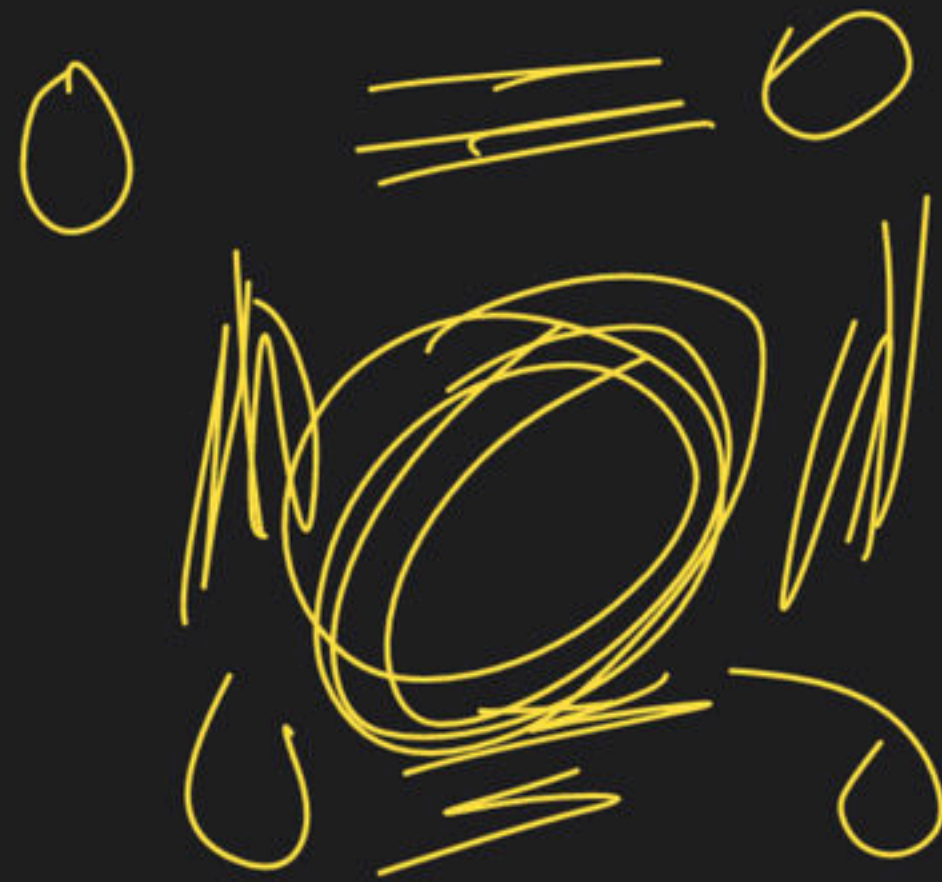
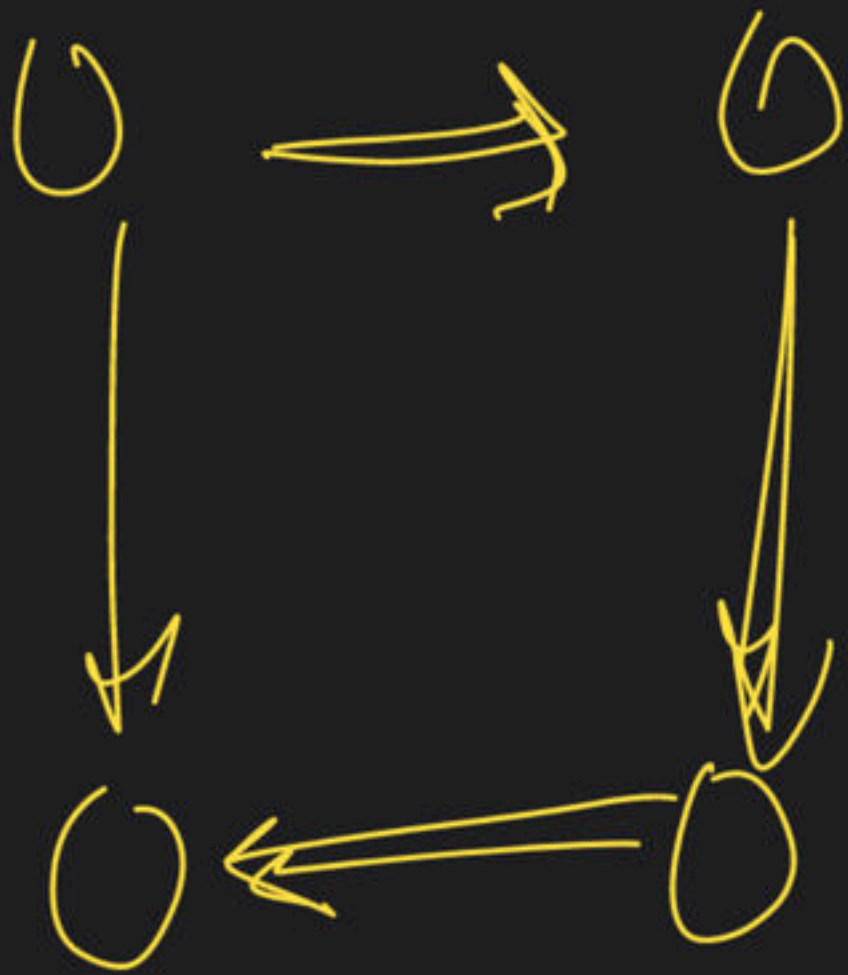
Directed Graph → BFS →  $\infty$   
→ DFS



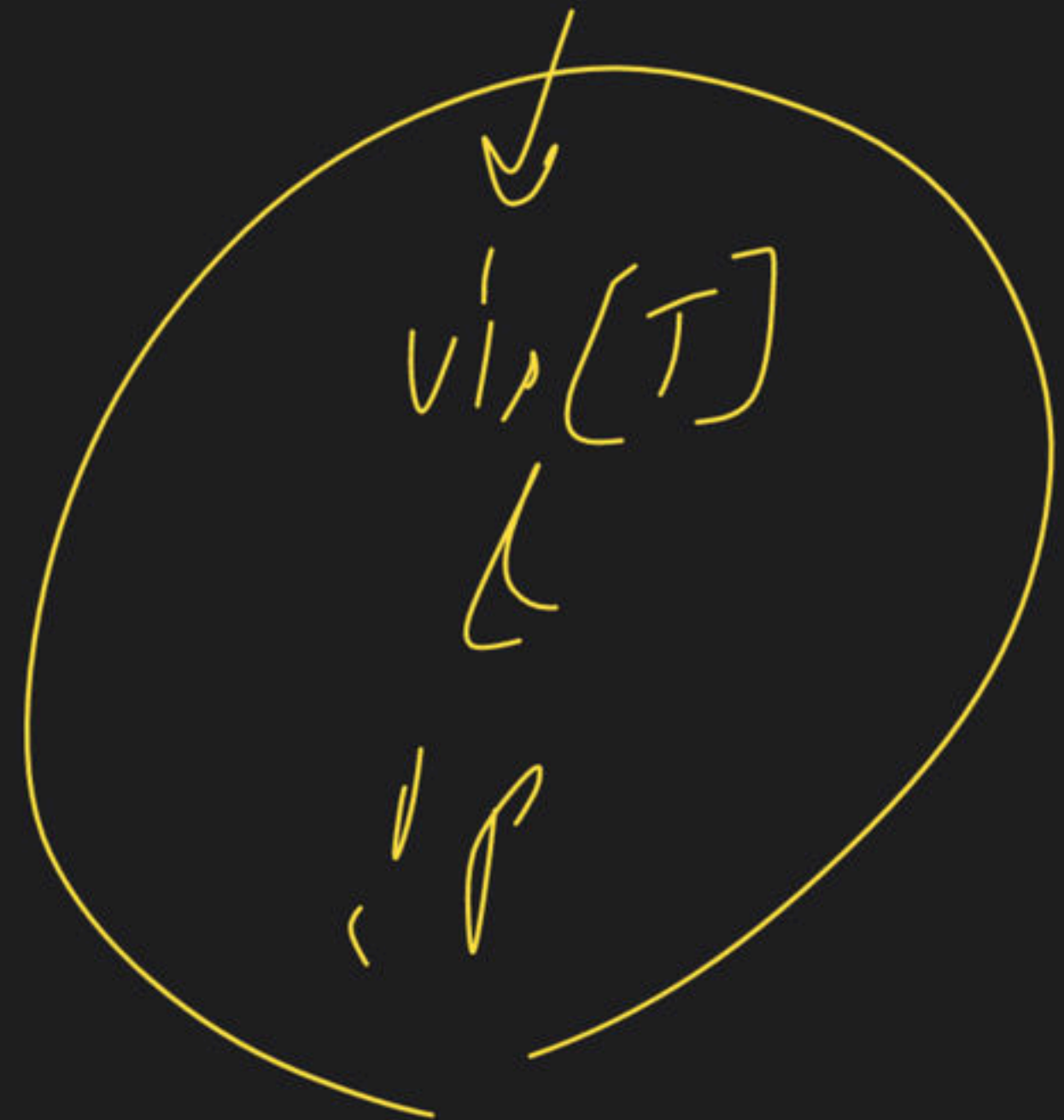
$$vis[nod] = 1$$

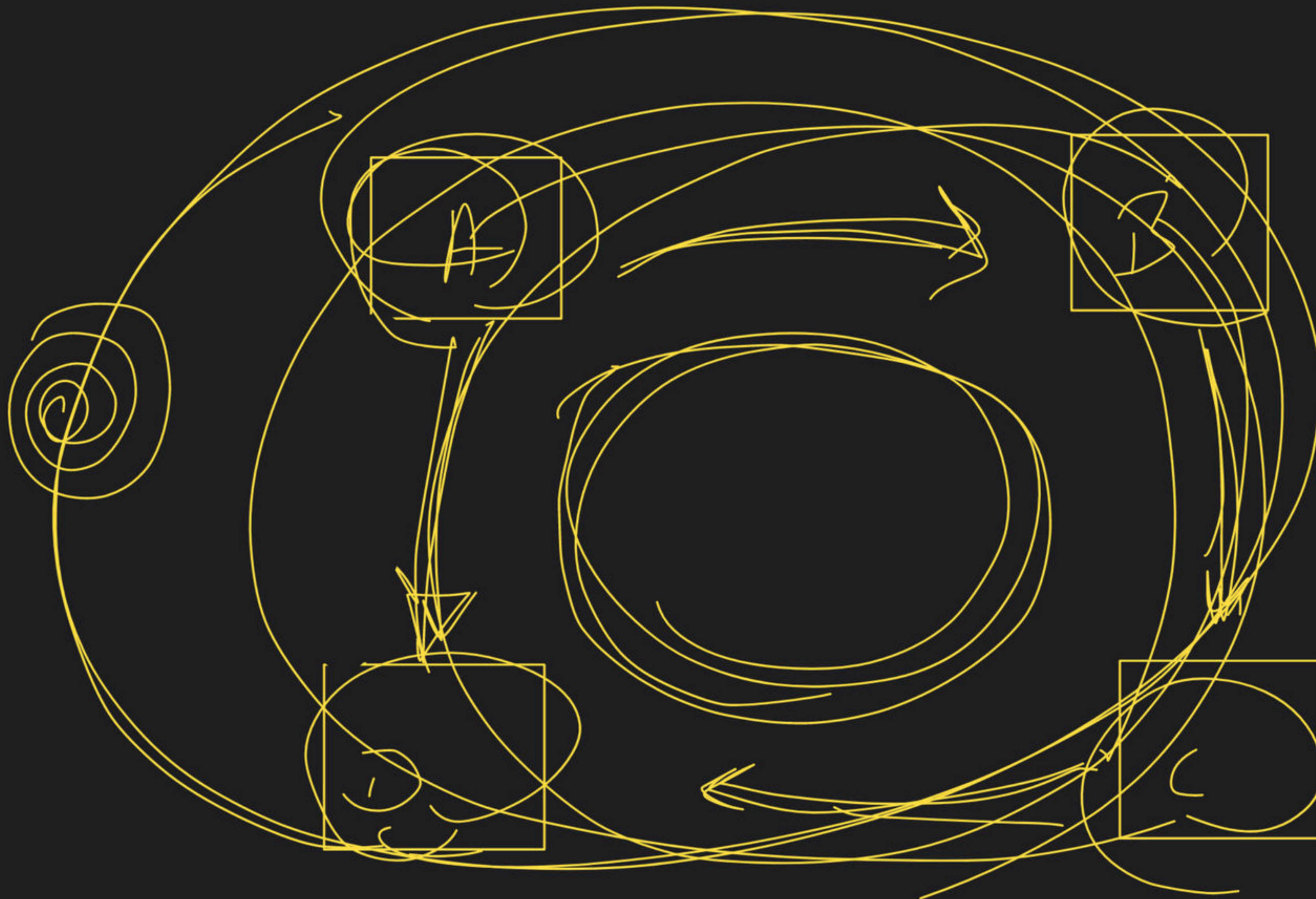


over path

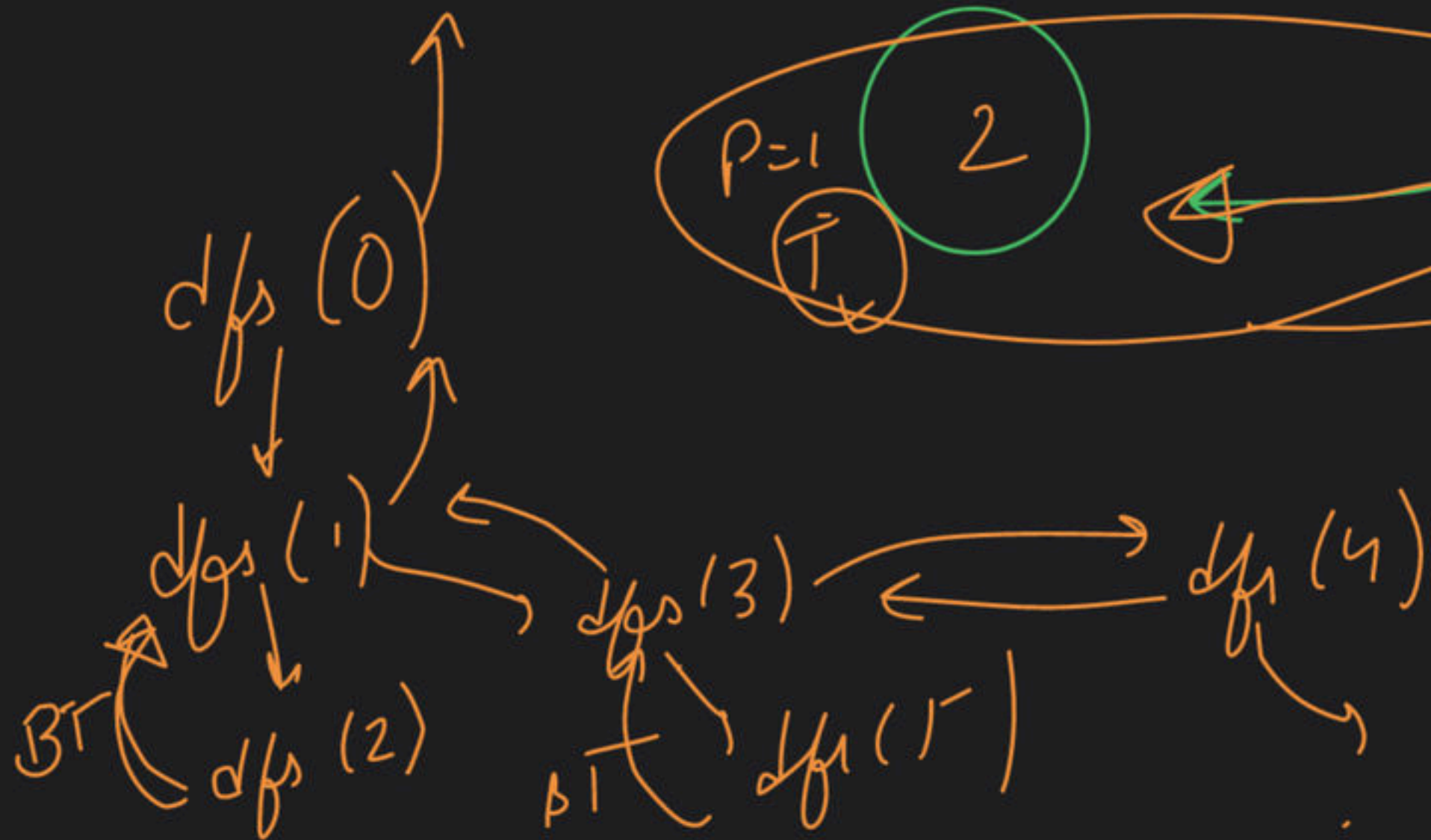


closed loop





A-B-C-D

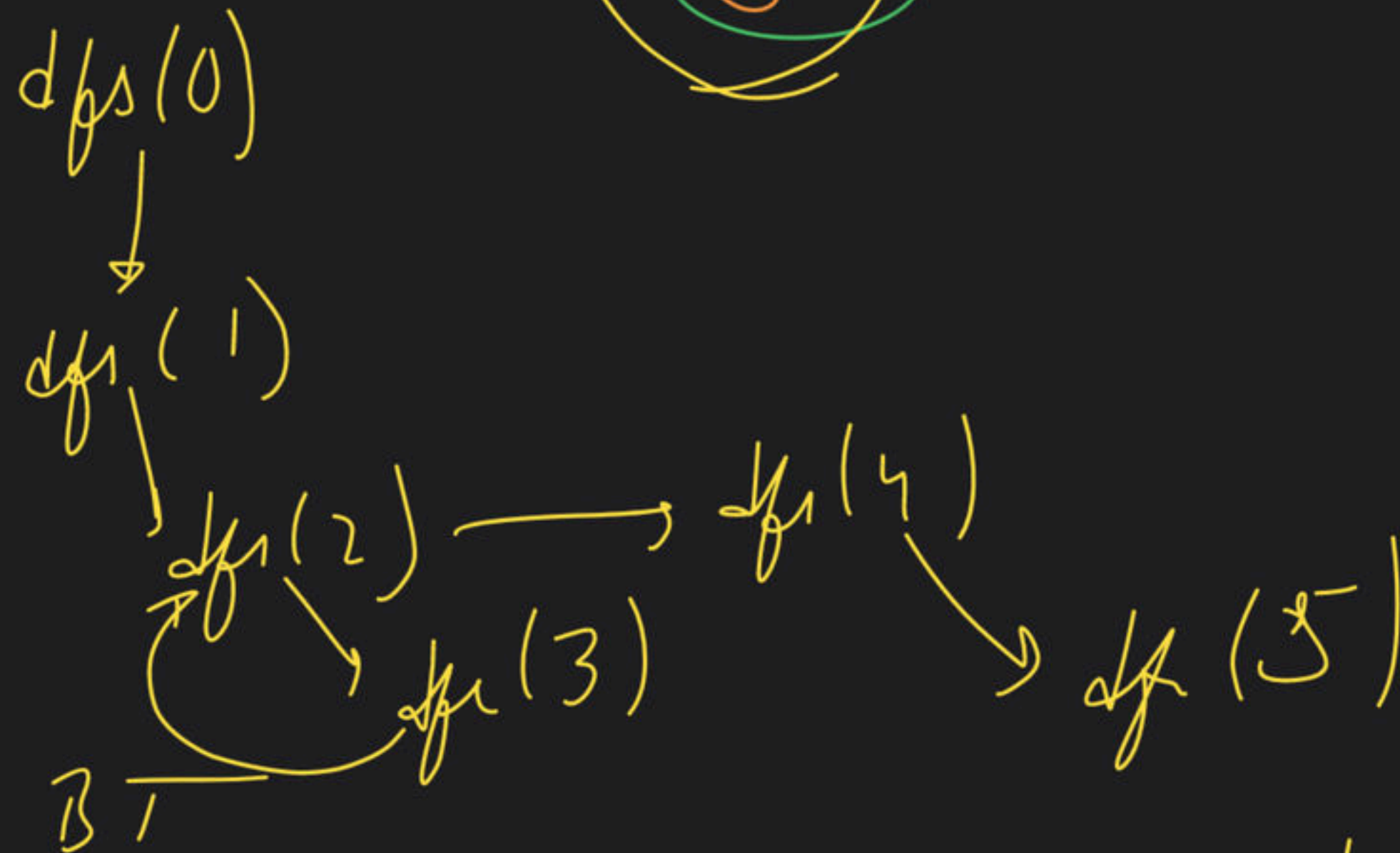
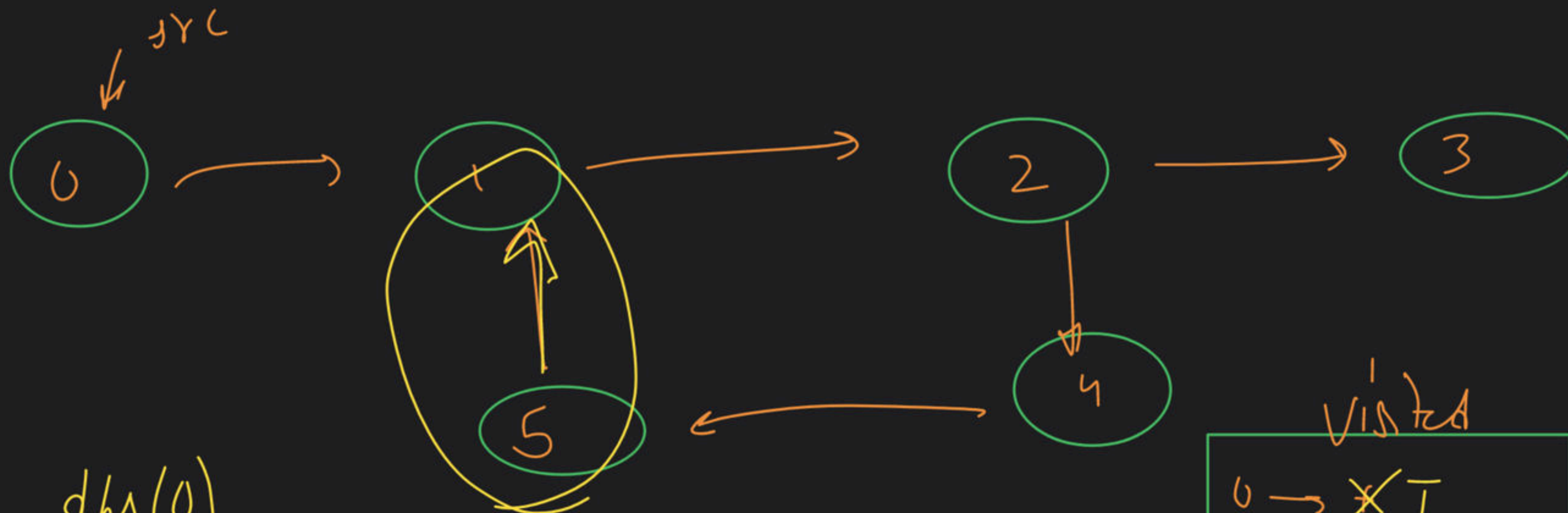


visited

|   |   |              |   |
|---|---|--------------|---|
| 0 | → | <del>X</del> | T |
| 1 | → | <del>X</del> | T |
| 2 | → | <del>X</del> | T |
| 3 | → | <del>X</del> | T |
| 4 | → | <del>X</del> | T |
| 5 | → | <del>X</del> | T |

dfsTracker

|   |   |              |   |
|---|---|--------------|---|
| 0 | → | <del>X</del> | F |
| 1 | → | <del>X</del> | F |
| 2 | → | <del>X</del> | F |
| 3 | → | <del>X</del> | F |
| 4 | → | <del>X</del> | F |
| 5 | → | <del>X</del> | F |



visited

|   |   |              |   |
|---|---|--------------|---|
| 0 | → | <del>X</del> | T |
| 1 | → | <del>X</del> | T |
| 2 | → | <del>X</del> | T |
| 3 | → | <del>X</del> | T |
| 4 | → | <del>X</del> | T |
| 5 | → | <del>X</del> | T |

dfsTracker

|   |   |              |   |
|---|---|--------------|---|
| 0 | → | <del>X</del> | T |
| 1 | → | <del>X</del> | T |
| 2 | → | <del>X</del> | T |
| 3 | → | <del>X</del> | F |
| 4 | → | <del>X</del> | T |
| 5 | → | <del>X</del> | T |

→ Cycle Present

























